Facilities Planning

LAYOUT

What is Facility Planning?

- **Facility planning** exercise determines how an activity's tangible fixed assets best support achieving the activity's objectives.
- In developing a layout for a system producing goods or services organisations seek to determine the best arrangement of facilities and equipment capable of satisfying anticipated demand at lowest cost.
- Facility planning is also know as layout planning, plant layout, facilities design.

Consequences of Poor Layout

- Congestion in aisles and storage areas and poor utilisation of space
- Excessive in process inventory and excessive work flow distances
- Continual production bottlenecks in some locations and simultaneous idle facilities elsewhere
- Skilled workers doing excessive unskilled work
- Long operation cycles and delivery delays
- Worker anxiety and strain
- Accidents or near accidents
- Obvious lack of production control

Need for Facility Planning

- **Effective facilities planning**:
  - can reduce the cost of material handling.
  - enables changes or adjustment to be made to the existing layout as a result of changes in demand and technology
  - provides easy access to equipment maintenance and repairs thus reducing downtimes and maintenance cost
  - minimizes losses in both money and manpower resulting from accident
  - provides a working environment that leads to a better utilization of human resources in the organization

Purposes of Re-layout

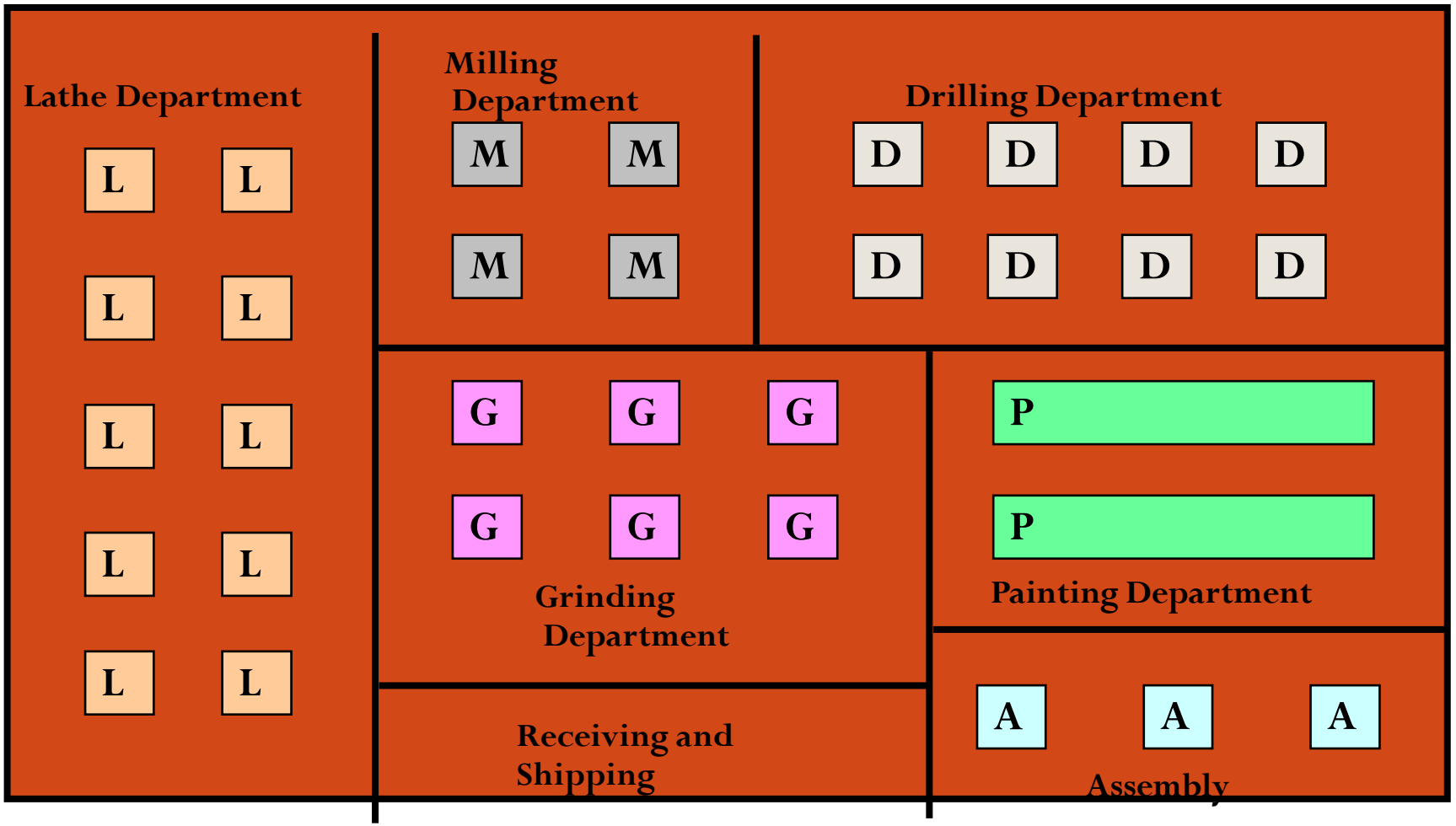
- To maximise the profitability or efficiency of operations
- Minimise safety or health hazards
- Freeing up bottleneck operations and
- Minimising interference, noise or distractions between different operational areas

Types of Layout

- Process Layout
- Product Layout
- Fixed Position Layout
- Cellular Layout
- Retail/Service Layout
- Warehouse Layout
- Office

Process Layout

- A process layout is the arrangement of **facilities** and **equipments** in groups according to function performed or in departments.



Advantages of Process Layout

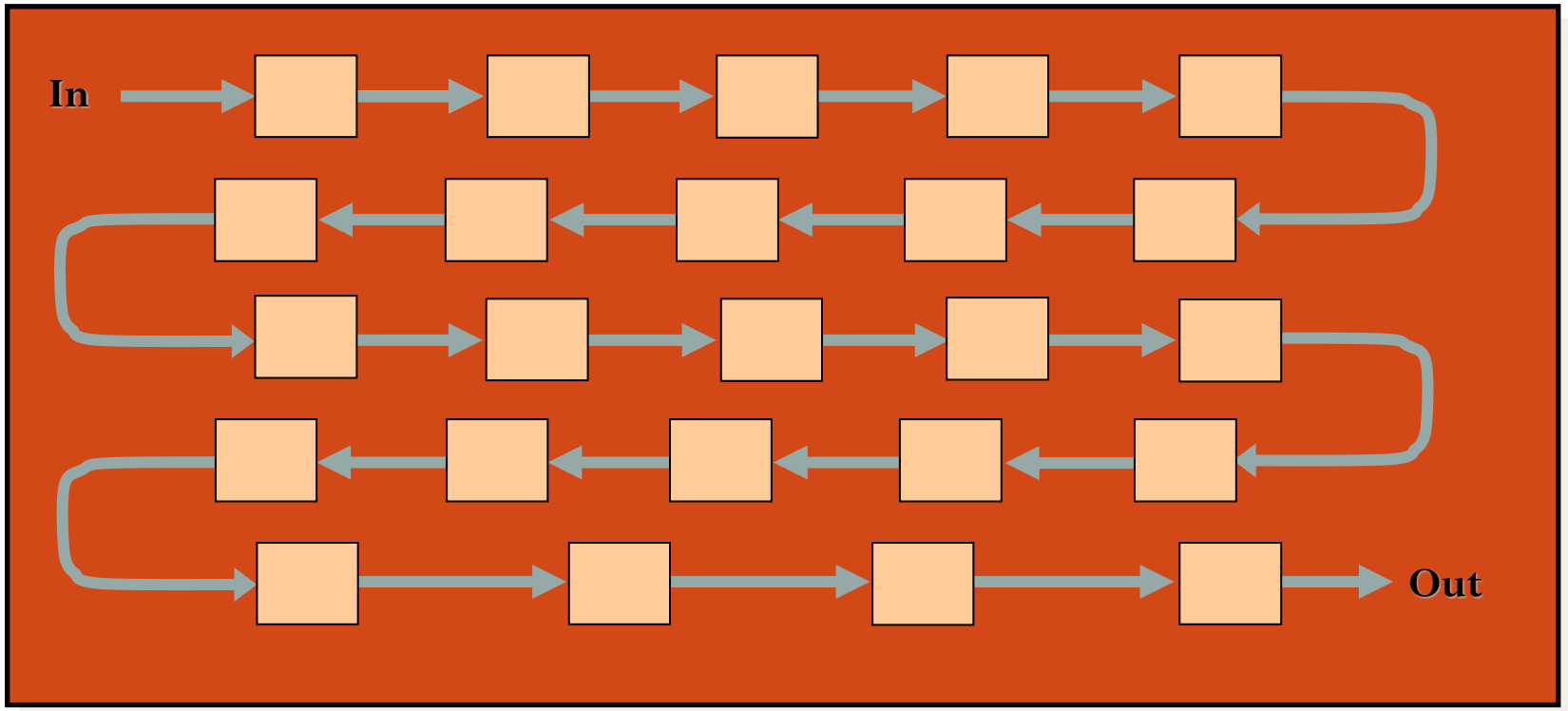
- Flexibility in equipment and labor assignment
- Manufacture of parts in small batches or job lots and the production of a wide variety of parts in different sizes or forms.
- Greater incentive for groups of workers to raise level of performance and greater possibility of group incentives.

Disadvantages of Process Layout

- Orders take more time and money to move through the system because of difficulty of scheduling, setups and material handling
- Higher total material handling cost
- Labor skill requirements are higher and this increases the required levels of training and experience
- Work-in-process inventories are higher because of higher imbalances in the production process

Product layout

- Is an arrangement of facilities and equipment in the same sequence as that of the operations needed to complete each unit of the product or the service offered.
- The production line is typically automated and uses such means as conveyors through a series of workstations.



Advantages of Product Layout

- Less duplication of equipment and hence lower total investment in equipment and training
- Low variable cost per unit because of high volume and standardized products
- Material handling costs are low
- Work-in-process inventories are also kept low
- Better and more efficient supervision possible through specialization
- Greater incentive to efficient individual workers

Disadvantages of Product Layout

- High volume is required because of high investment needed to set up the process
- Work stoppage at any one point ties up the whole operation
- Less flexibility of production

Comparison of Product and Process Layouts

	Product	Process
Description	Sequential arrangement of activities	Functional grouping of activities
Type of process	Continuous, mass production, mainly assembly	Intermittent, job shop, batch production, mainly fabrication
Product	Standardized, made to stock	Varied, made to order
Demand	Stable	Fluctuating
Volume	High	Low
Equipment	Special purpose	General purpose

Comparison of Product and Process Layouts

	Product	Process
Workers	Limited skills	Varied skills
Inventory	Low in-process, high finished goods	High in-process, low finished goods
Storage space	Small	Large
Material handling	Fixed path (conveyor)	Variable path (forklift)
Aisles	Narrow	Wide
Scheduling	Part of balancing	Dynamic
Layout decision	Line balancing	Machine location
Goal	Equalize work at each station	Minimize material handling cost
Advantage	Efficiency	Flexibility

Developing the Process Layout

1. Analyse the product or products to be produced
   - Complete design drawings
   - The parts list which established those parts to be manufactured and/or purchased and which must be provided for the general plant area under consideration
   - Assembly charts indication the sequence by which the parts are combined into assemblies

Developing the Process Layout(cont.)

2. Determine the process required to manufacture the product

- Route sheets and operations sheets must be obtained or developed for each manufactured part or assembly
- Operation process charts are prepared following the preparation of the route sheets to provide a means of combining the assembly charts and route sheet data in a single form

3. Prepare layout planning chart which includes:

- Flow process showing all operations, moves, storages and inspections
- Standard times for each operation obtained from time study or pre-determined standards

Developing Process Layout(cont)

- Machine selection
- Manpower requirements for the production activity
- Machine balance and manpower balance
- Material handling load, methods and equipment requirements

4. Determine the work stations

- Layout must be developed taking into consideration machine, operator, materials and service area requirement

Developing the process layout(contd)

5. Analyse storage area requirement
6. Establish minimum aisle widths
7. Establish office requirements
8. Consider personnel facilities and services
9. Survey plant services
10. Provide for further expansion

Process Oriented Layout

- The most common tactic is to arrange departments or work centres in the most **economical locations** to **minimize material handling cost.**
- - This includes placing departments with **large inter-departmental** flow of parts or people next to one another.

Process Oriented Layout

The objective can be represented as:

$$= \sum_{i=1}^{n} \sum_{i=1}^{n} XijCij$$

n = total number of workstations or departments

I,j = individual workstations or departments

X_{ij} = number of loads moved from one department `i` to department `j`

C_{ij} = Cost to move a load between departments `i` and `j`

Process Layout: An Example

The Walters Company`s management wants to arrange the six (6) departments of its factory in a way that will minimize inter-departmental material handling cost. They make an initial assumption (to simplify the problem) that each department is 20 X 20 feet long and that the handling is 60ft long and 40 ft wide. The cost of moving one load to between adjacent departments is estimated to be $1 and between non-adjacent departments is $2.

The process flow layout procedure you follow are in six (6) steps

Steps

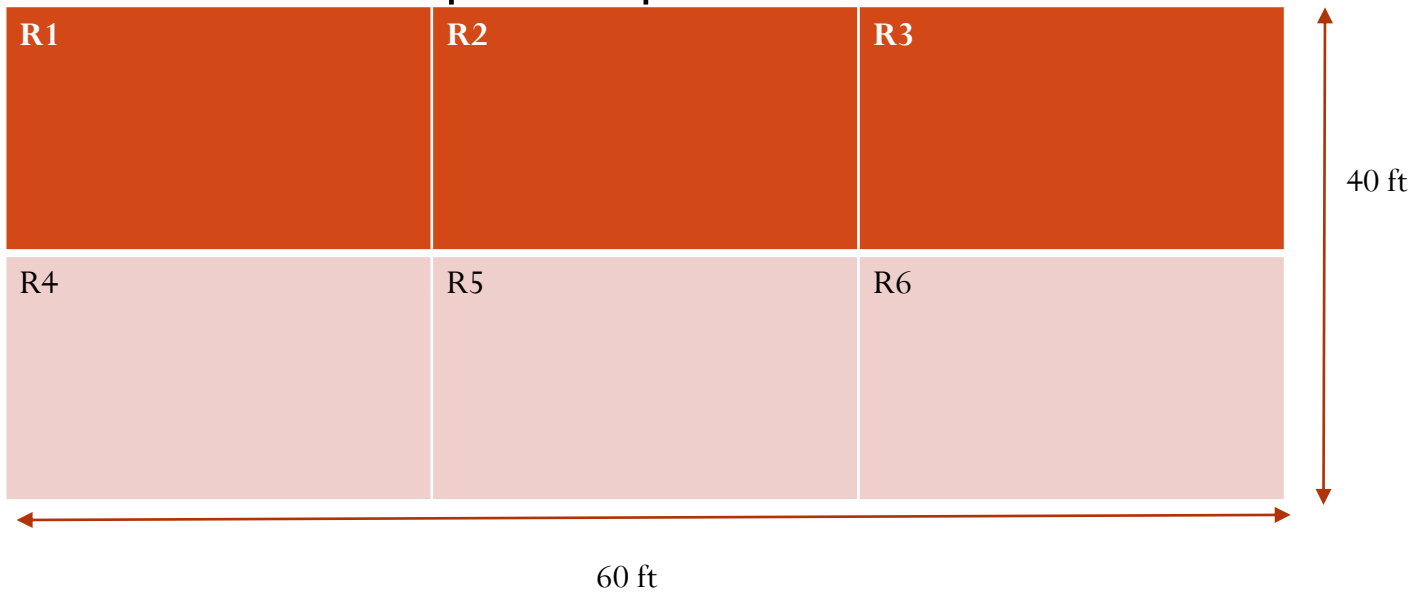
- Step 1

Construct a from-to-matrix showing the flow of the parts or material from department to department.

	1	2	3	4	5	6
1		**50**	**100**	**0**	**0**	**20**
2			30	50	10	0
3				20	0	100
4					60	0
5						0
6						

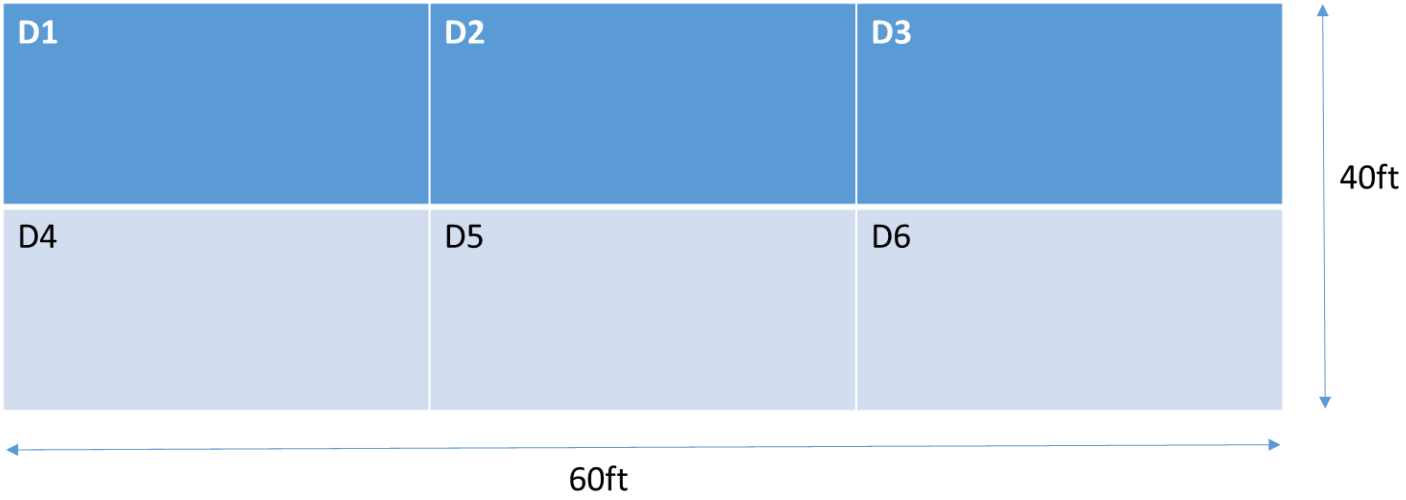
- Step 2

Determine the space requirement



- Step 3

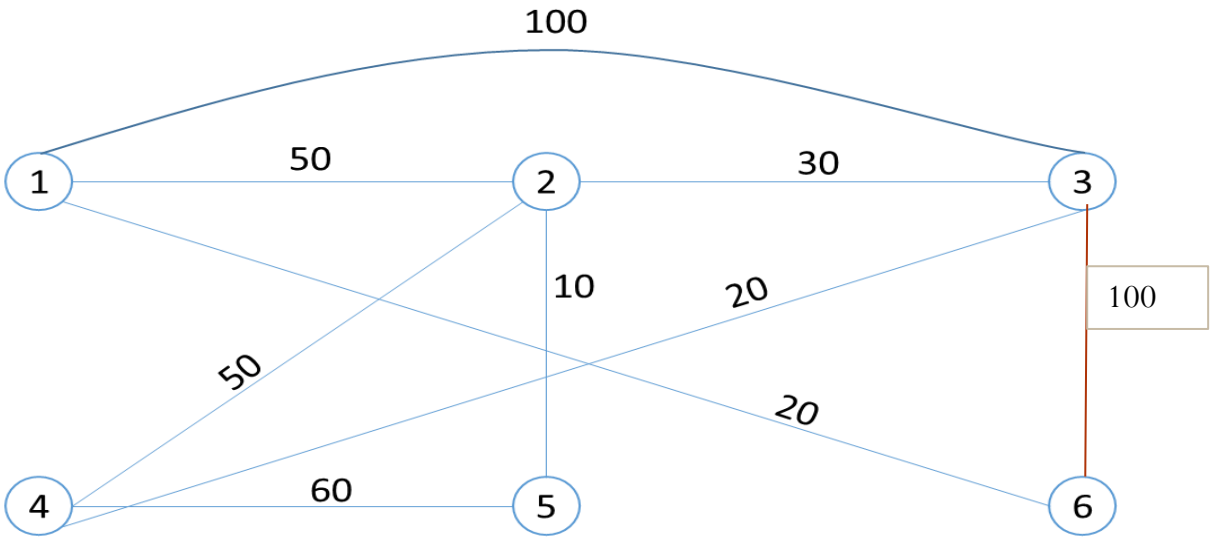
Develop an initial schematic diagram showing the sequence of the departments through which parts will have to move. Try and place departments with heavy load between them.



- Step 4

Determine the cost of the layout by using the material handling cost equation.

$$\text{Cost} = \sum_{i=1}^{n}\sum_{i=1}^{n} XijCij$$



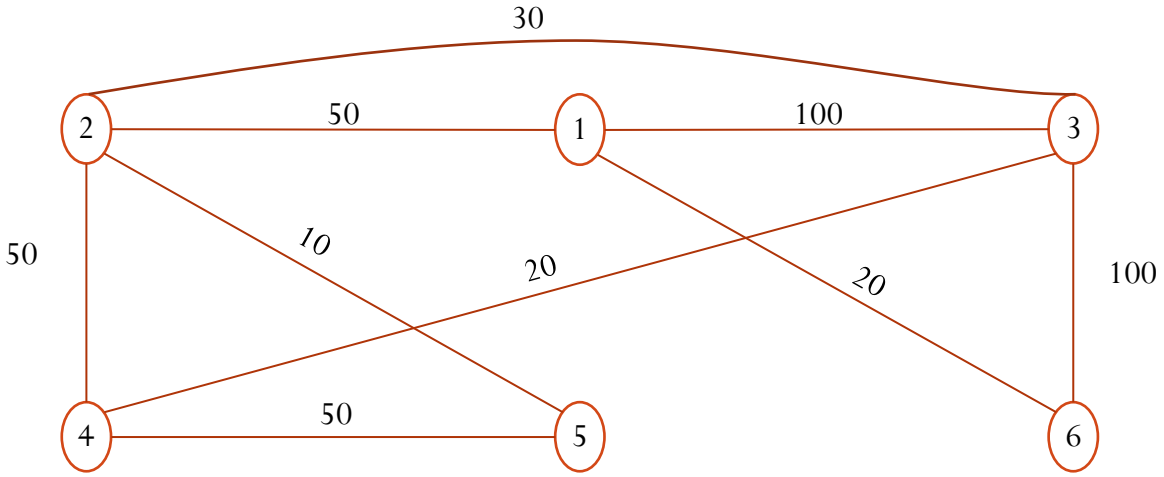
Cost = (1 & 2) + (1 & 3) + (1 & 6) + (2 & 3) + (2 & 4) + (2 & 5) + (4 & 3) + (3 & 6) + (4 & 5)

=(50 + 200 + 40 + 30 + 50 + 10 + 40 + 100 + 50)

=$570

- Step 5

By trial and error(or by sophisticated computer programs) try to improve this layout to establish a reasonably good arrangement of department



2	1	3
4	5	6

Associated cost

= (1 & 2) + (1 & 3) + (1 & 6) + (2 & 3) + (2 & 4) + (2 & 5) + (3 & 4) + (3 & 6) + (4 & 5)

= $480

The problem may not be over because you could have as many as the factorial of the number of departments. For the problem above you could have as many as 6! arrangements.

6! = 6 x 5 x 4 x 3 x 2 x 1

= 720 arrangements

- Step 6

Prepare a detailed plan considering space or size requirement of each department.

i.e. Fitting the departments to the shape of the building and its non-movable areas(such as loading dock , washroom etc.)

Often the six steps involves making sure that the final plan can be accommodated by the electrical system, floor loads, aesthetics and other factors.

Travel Chart

From Departments	To Departments					
	A	B	C	D	E	F
A - Assembly	-	50	100	50	80	130
B -Painting	85	-	50	-	120	70
C -Machine Shop	80	60	-	140	100	50
D -Receiving	85	180	80	-	120	220
E -Shipping	-	-	-	-	-	-
F -Testing	70	90	20	-	140	-

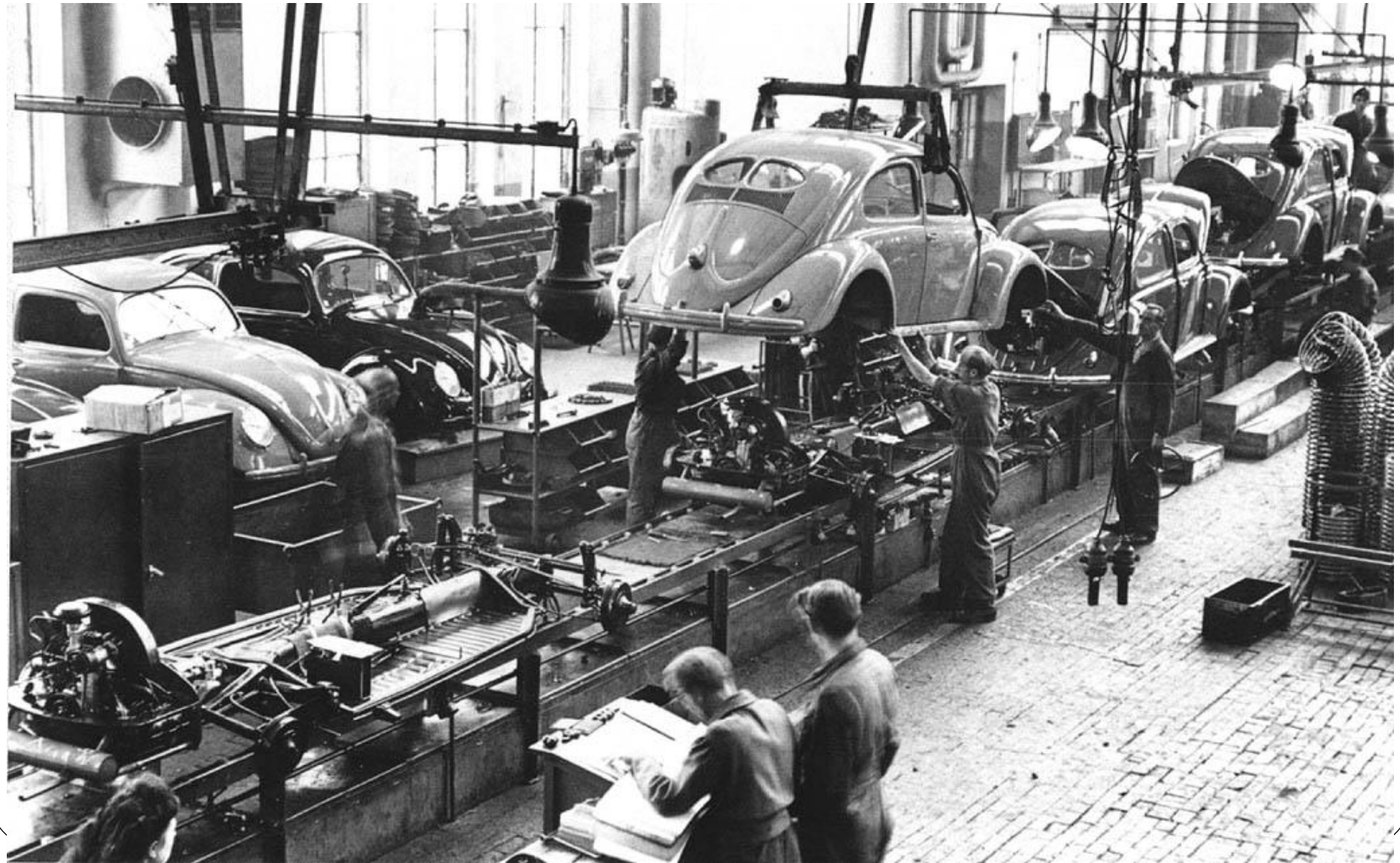
Conventional Approach for Developing Product Layout

- Product layouts are suitable for mass production of discrete items
- Products layout are used to achieve smooth flow of large volume of highly standardised products that require repetitive processing operations
- The main issue in design of product layout is line balancing
  - **The process of deciding how to assign tasks to work stations on the line is referred to as line balancing**
- The objective of line balancing is to obtain task groupings that represent approximately equal time requirement.

Conventional Approach for Developing Product Layout(cont)

- Perfect balance would lead to smooth flow of work
- It is difficult to achieve perfect balancing as a result of inability to obtain task groupings that have same durations
- The cycle time which is the amount of time each work station has to complete its set of task before the product moves to the next station, determines the output rate per line.

Assembly Line



Assembly Line

An **assembly line** is a manufacturing process (most of the time called a *progressive assembly*) in which parts are added to a product in a sequential manner to create a finished product.

It is much faster way of producing a product.

➢Assembly lines work on the principle of division of labour propounded by Adam Smith.

- Break the production process of product into elemental task

-Group tasks together

-Set up the workers, tools or machines, and parts in a sequential order

-Each worker typically performs one simple operation.

-The motion of workers is minimized to the extent possible.

All parts or assemblies are transported through different work centers using either conveyors or motorized vehicles

Typical Delay Scenarios

1. **Flow-blocking delay**: Work-in-progress cannot be released because the following workstations is occupied
2. **Lack of work delay**: Work has been completed at one station but new job has not arrived.

- These are indications of an unbalanced line.
- The assembly line technique builds on the operations chart and the plant rate.
- The objective is to give each operator as close to the same amount of work as possible.
- This can only be done by breaking the work into basic motions to do every single piece of work and reassembling into jobs of near equal time value.

Assembly Line Balancing

The technique of ensuring that each workstation has a near equal time is what is known as assembly line balancing.

Purpose:

Equalise workload among the assemblers

Identify bottleneck operations

Establish the speed of the assembly line

Determine the number of work stations

Determine the labour cost of assembly and packout

Establish the percentage workload of each operator

Assist in plant layout

Reduce production cost

Assembly Line Balancing

Cycle time: that is the time each product is available at each workstation

$$\text{Cycle time} = \frac{\text{Production time available per day}}{\text{Demand per day}}$$

$$\text{Cycle time(d)} = \frac{\text{Production time available per day}}{\text{Production rate per day}}$$

$$\text{Theoretical minimum number of workstations} = \frac{\text{Summation of elemental time}}{\text{Cycle time}}$$

$$\text{Efficiency of the line} = \frac{\text{Summation of elemental time}}{\text{Actual number of workstation} * \text{Cycle time}}$$

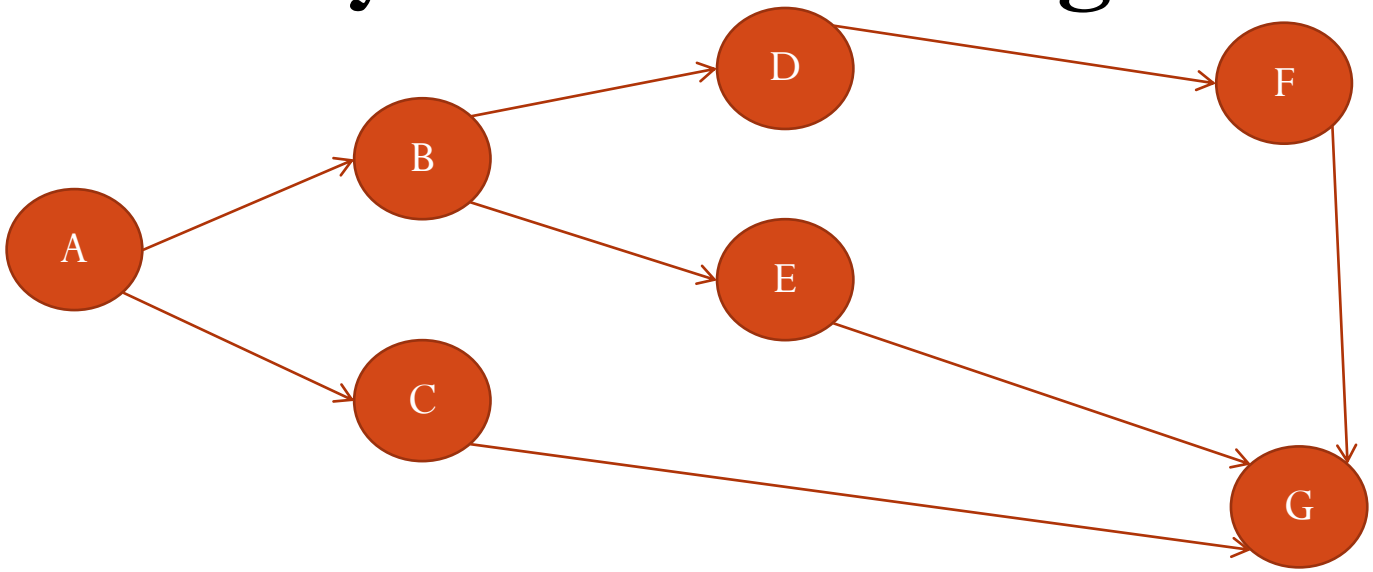
Assembly Line Balancing: An Example

A	-	4
B	A	2
C	A	1
D	B	1
E	B	5
F	D	3
G	CEF	4

The Production Manager is tasked to produce 120 units of the product in a 10hr a day total available production time.

1. Determine the theoretical minimum number of workstations?
2. What is the actual number of workstations?
3. What is the efficiency of the line?

Assembly Line Balancing: An Example



Cycle time $= \frac{10 * 60}{120}$

$= 5$ minutes

Theoretical minimum number of workstations is $= \frac{4+2+1+1+5+3+4}{5} = \frac{20}{5} = 4$

Assembly Line Balancing: An Example

Rule of thumb: Always assign the longest time task first

Work station 1

Task	Time used	Time remaining
A	4	1
C	1	0

Work station 2

Task	Time used	Time remaining
B	2	3
D	1	2

Work station 3

Task	Time used	Time remaining
E	5	0

Work station 4

Task	Time used	Time remaining
F	3	2

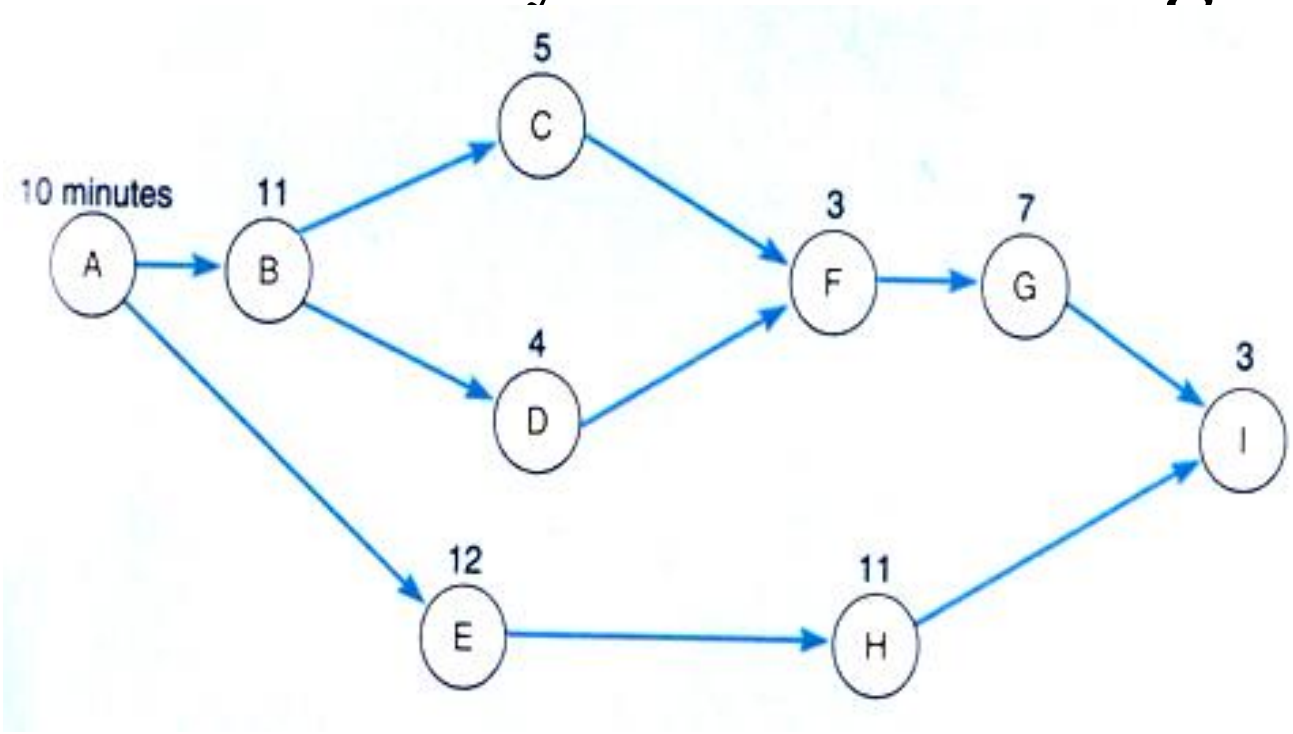
Work station 5

Task	Time used	Time remaining
G	4	1

Efficiency of the line $= \frac{\text{Summation of time at each workstation}}{\text{Actual number of workstation} * \text{Cycle time}}$

$= \frac{20}{5*5} = 80\%$

Assembly Line Balancing: Example 2



Assume daily production quantity of 122
and a 10 hour available production time.

1. What is the cycle time
2. What is the theoretical minimum number of work stations?
3. What is the actual number of work stations?
4. What is the efficiency of the assembly line?

Fixed-Position Layouts

- Typical of projects in which product produced is too fragile, bulky, or heavy to move
- Equipment, workers, materials, other resources brought to the site
- Low equipment utilization
- Highly skilled labor
- Typically low fixed cost
- Often high variable costs



Material Handling

Material handling: the movement, storage, control and protection of materials, goods and products throughout the process of manufacturing, distribution, consumption and disposal

Objectives of Material Handling

Increase capacity

Improve working condition

Improved customer service

Increased equipment and space utilisation

Reduced cost

Automatic Material Handling System



Conveyor



AGV

Material Handling: Assignment

1. Define what a material handling systems is.
2. What are the main categories of material handling systems?
3. Identify two material handing systems, belonging to different categories and write about them:
    1. Name, application and uses,
    2. Parts,
    3. Operation
    4. Pictures and diagrams

Total 10 marks

Deadline for submission 25th of March, 2015